RAG-NIMBF: A Non-Invasive, Modular Benchmarking Framework for Automated RAG Configuration Analysis

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Problem

Retrieval-augmented generation (RAG) systems expose a large configuration space. Chunking, overlap, embedding model, vector database, retrieval strategy, reranking, prompt template, generator model, and evaluator choice can all move quality and runtime in different directions.

Research goal: make RAG configuration choices measurable, repeatable, and comparable without rewriting the target pipeline for each experiment.

Framework Contribution

- Environment-driven benchmark grids for local and OpenAI-compatible model providers.
- Dataset adapters normalize QA corpora into one benchmark record shape.
- Reusable retrieval core supports Chroma and LanceDB vector stores, similarity search, MMR, optional HyDE, and cross-encoder reranking.
- Generation layer captures answer text, time to first token, throughput, and optional GPU statistics.
- Reporting exports JSON, CSV, Markdown, static plots, MLflow runs, and optional tracing artifacts.

Pipeline

Configuration grid
models, chunks, prompts

Dataset adapter
QA record shape

Chunking + vector-store cache
reusable index for each corpus/configuration

Retrieval, optional reranking, prompt assembly
similarity/MMR contexts become model input

Generation
answer text, TTFT, throughput, GPU stats

RAGAS + custom IR/NLG metrics
faithfulness, similarity, retrieval quality, runtime

Artifacts
JSON, CSV, Markdown, plots, MLflow, traces

Experiment Space

Dimension	Current values in project
Chunking	recursive, semantic, text, token, markdown
Chunk size	500, 1000, 1500 and variants
Overlap	100, 200, 350 and variants
Retrieval	similarity, MMR
Reranking	none, MiniLM cross-encoder
Embedding	`nomic-embed-text:latest` snapshot
Datasets	SQuAD / WikiQA-style QA adapters
Models	Qwen3, Qwen3.5, GPT-OSS local runs

Reproducibility Controls

- **Config grid:** environment-defined combinations generate every run.
- **Run folders:** each benchmark writes to `results/runN/`.
- **Vector cache:** corpus fingerprints avoid accidental re-embedding.
- **Stage timings:** indexing, retrieval, generation, and evaluation are timed.
- **Artifact trail:** JSON, CSV, plots, MLflow, and traces preserve evidence.

Observed Answers

- **Prompting:** detailed prompts improve mean faithfulness (0.885 vs. 0.827); the best faithfulness run is detailed (0.955).
- **Retrieval:** similarity is stronger for faithfulness (0.870 vs. 0.843), while MMR improves nDCG@5 (0.796 vs. 0.715) and recall@5 (0.847 vs. 0.554).
- **Chunking:** chunk size and overlap have smaller faithfulness effects than prompt or retrieval choice (500: 0.854, 1000: 0.857; overlap 100: 0.854, 200: 0.858).
- **Winner:** no single configuration dominates: faithfulness, BERT-F1, nDCG@5, and recall@5 select different runs.

Evaluation Lenses

- **LLM-judged quality:** faithfulness, context recall, semantic similarity, and related RAGAS fields where available.
- **Retrieval quality:** hit@k, nDCG@k, recall@k, context relevance, with heuristic or gold-document modes.
- **Answer similarity:** ROUGE-L, BLEU, METEOR, and optional BERTScore precision, recall, F1.
- **Performance:** total runtime, stage timings, time to first token, tokens per second, and GPU usage when available.

Prompt Templates

Concise Prompt

```
Answer the question using ONLY the provided context.
RULES:
- Give a short, direct, factual answer - one sentence maximum.
- Do NOT explain your reasoning, add context, or use markdown formatting.
- Do NOT say 'Based on the context' or similar preamble.
- If the context does not contain the answer, say 'I cannot answer from the provided context.'
Examples:
Q: What circuit court is Maryland? → The Circuit Courts of Maryland are the state trial courts of general jurisdiction.
Q: How many members are in the House of Representatives? → The total number of voting representatives is fixed at 435.
Q: Who wrote the music for Star Wars? → John Williams.
```

Detailed Prompt

```
Answer the question based only on the provided context. Provide a clear, complete answer in full sentences. Include relevant calculations or reasoning steps when applicable. Do not start your answer with phrases like 'Based on the provided context' or 'Based on the given context'. Answer directly.
```

```
Human template for both: Context:\n{context}\n\nQuestion: {question}\n\nAnswer:
```

Current Results Snapshot

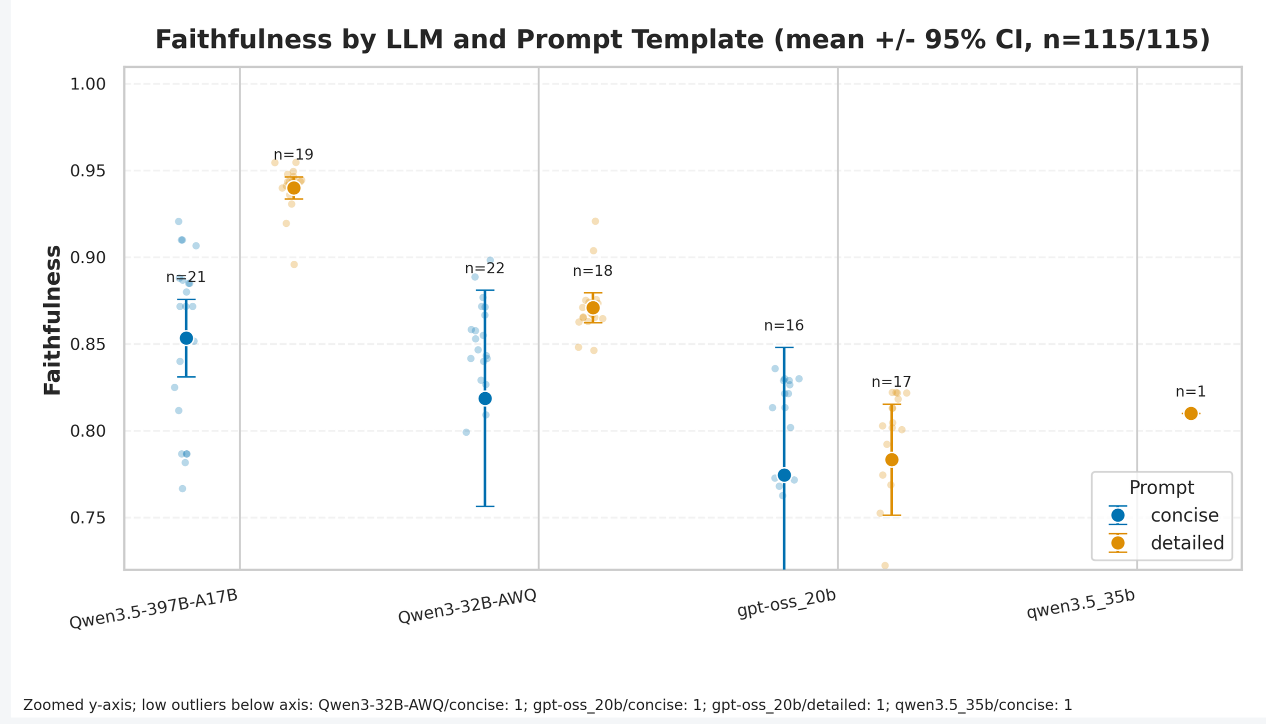
The current state of the project covers 128 benchmarked configurations, more are planned for future work. The table below highlights the strongest configuration observed for each of the three main evaluation lenses, demonstrating that different configurations can excel in different metrics, underscoring the importance of multi-metric evaluation in RAG benchmarking.

Observed strongest configuration	Faith.	BERT-F1	nDCG@5
Qwen3.5-397B-A17B, detailed prompt, recursive chunks	0.955	–	–
Qwen/Qwen3-32B-AWQ, concise prompt, semantic chunks, MMR	0.871	0.969	1.000
Qwen3.5-397B-A17B, detailed prompt, semantic chunks	0.948	–	–

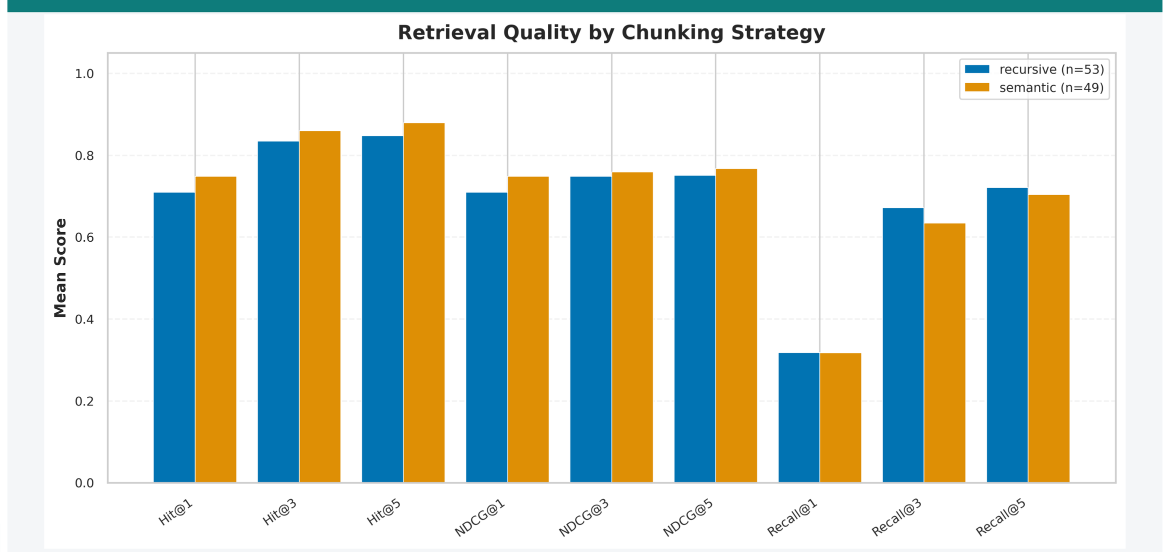
Interpretation: the best run depends on the metric. Strong faithfulness, strong retrieval ranking, and strong lexical/semantic answer similarity do not always select the same configuration.

Quality Ranking

Prompt and Model Effects

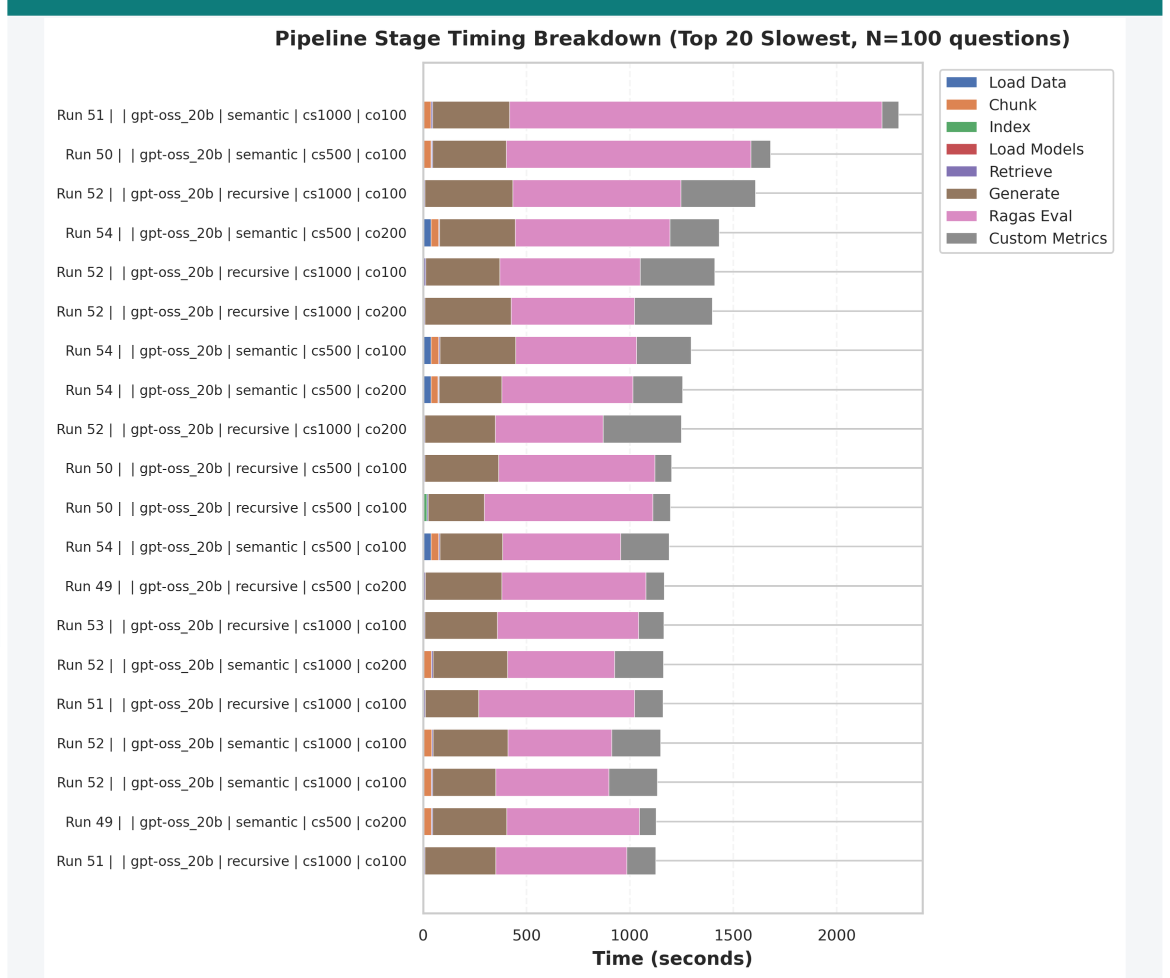


Retrieval Trade-Offs



MMR can improve diversity-sensitive retrieval metrics in some runs, while plain similarity can preserve higher faithfulness in others. Reranking helps in selected settings, but adds latency and should be judged jointly with answer quality.

Runtime and Engineering



- Persisted vector-store collections avoid unnecessary re-embedding.
- Stage-level timings separate indexing, retrieval, generation, and evaluation bottlenecks.
- Sequential execution matches local GPU constraints and reduces resource contention.

Limitations

- Current results are local, hardware-dependent, and partly duplicated by reruns.
- SQuAD/WikiQA-style corpora may not transfer to domain RAG workloads.
- LLM evaluator reliability, stochastic generation, and provider changes remain validity threats.
- Manual matrix entries must be reconciled with generated result files before publication.

Takeaways for Practitioners

1. Benchmark complete RAG configurations, not isolated components.
2. Report multiple metrics because each metric rewards different behavior.
3. Store machine-readable artifacts for every run.
4. Treat prompt template and retrieval strategy as first-class tuning variables.
5. Keep limitations visible when using LLM-judged metrics.

Measured Outputs

Lens	Captured outputs
Answer	faithfulness, recall, semantic similarity
Retrieval	hit@k, nDCG@k, recall@k, relevance
Runtime	total time, stage timings, TTFT, tokens/s
Artifacts	JSON, CSV, Markdown, plots, MLflow, traces

References

Lewis et al. (2020), Es et al. (2023), Rajpurkar et al. (2016), Lin (2004), Zhang et al. (2020).

scan for code